Quantification: 872_wt_hp_z03
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	9
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 9)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1305.5	1243.5	612.612	px^2
major_axis_length_px	lacuna shape	66.2315	68.6165	14.1596	px
minor_axis_length_px	lacuna shape	25.7426	25.8032	9.8277	px
aspect_ratio	lacuna shape	2.8467	2.4831	1.1271	unitless
eccentricity	lacuna shape	0.9082	0.9153	0.0591	unitless
solidity	lacuna shape	0.8777	0.8936	0.0449	unitless
roots_count	headline	7.2222	6	3.5978	count
ring_length_r30_px	headline	319.111	275	119.867	px
ring_length_r60_px	reported	846.778	697	293.916	px
owned_length_px	ownership-dependent	1606.86	462.375	1692.73	px
edge_count	ownership-dependent	99.4444	25	107.556	count
mean_edge_length_px	ownership-dependent	16.9473	16.2445	2.21	px
perimeter_px	normalised	169.531	180.267	56.9961	px
ring_area_r30_px2	normalised	7224.22	7666	1379.94	px^2
ring_area_r60_px2	normalised	19695.8	20442	2753.88	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0444293	0.0356979	0.0145984	px^-1
ring_density_r60	normalised	0.0430795	0.0407336	0.0132482	px^-1
roots_per_100px_perimeter	normalised	4.4559	4.4379	1.8568	per 100 px
ring_attached_length_r30_px	network v2	278.222	237	132.54	px
ring_attached_length_r60_px	network v2	588	465	389.837	px
ring_length_w_r30_px	network v2	364.951	316.907	140.028	px
ring_length_w_r60_px	network v2	962.802	796.453	340.028	px
sholl_crossings_r10	network v2	6.4444	6	2.0069	count
sholl_crossings_r20	network v2	8	7	3.6056	count
sholl_crossings_r30	network v2	11.7778	11	4.1767	count
loop_count	loops, report only	11.2222	2	13.2832	count
loop_edge_share	loops, report only	0.2648	0.28	0.2162	share
ring_width_mean_r30_px	width	7.1616	6.7744	0.8905	px

Per field
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measure	value	units
analysed_area_px2	1.03214e+06	px^2
total_skeleton_length_px	27710	px
canalicular_length_density_per_px	0.0268472	px^-1
junction_count	742	count
junction_density_per_px2	0.0007189	px^-2
lacuna_count	12	count

lacunae_per_px2	1.163e-05	px^-2
skeleton_component_count	401	count
mean_component_length_px	69.1022	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0301605	px^-1
field_density_without_flagged_per_px	0.027309	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	498	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0268472	px^-1
field_length_density_with_unattached_per_px	0.0273297	px^-1
loop_count	134	count
loop_edge_share	0.2652	share
edge_count	1859	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.2334	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	25079	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	794	69.6134	15.6156	4.4579	0.9745	0.834	-1.442	210.909	115.354	False
2	420	36.8904	15.1005	2.443	0.9124	0.8955	-0.9718	230.283	755.805	False
3	1078	83.4971	17.8088	4.6885	0.977	0.8021	-1.3939	282.36	422.258	False
4	2506	64.9879	57.5952	1.1284	0.4632	0.8735	0.3596	332.631	998.485	True
5	2428	74.728	42.3101	1.7662	0.8243	0.9324	-1.5637	398.271	285.478	False
6	2726	99.126	35.5862	2.7855	0.9333	0.9172	-1.4054	396.683	51.522	True
7	1386	67.6197	26.7997	2.5232	0.9181	0.9053	-1.5154	396.681	828.755	False
8	1506	75.3481	26.717	2.8202	0.935	0.8917	-1.4312	511.238	191.668	False
9	1101	57.7118	24.8895	2.3187	0.9022	0.9137	-1.3389	569.522	707.45	False
10	568	28.9763	25.6237	1.1308	0.4669	0.9578	-0.8322	590.484	11.2324	True
11	1731	64.4432	36.6995	1.756	0.822	0.8469	-1.4227	651.578	232.505	False
12								270.839	634.653	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	242	532	253.024	12	21.0854	6.4399
2	6	192	581	192.066	13	14.7743	5.808
3	5	230	622	291.966	17	17.1744	6.5575
4	6	272	717	1801.46	99	18.1966	6.2067
5	9	312	904	374.492	25	14.9797	7.7029
6	4	197	360	124.083	5	24.8167	5.7427
7	8	407	1143	2334.7	142	16.4415	6.7143

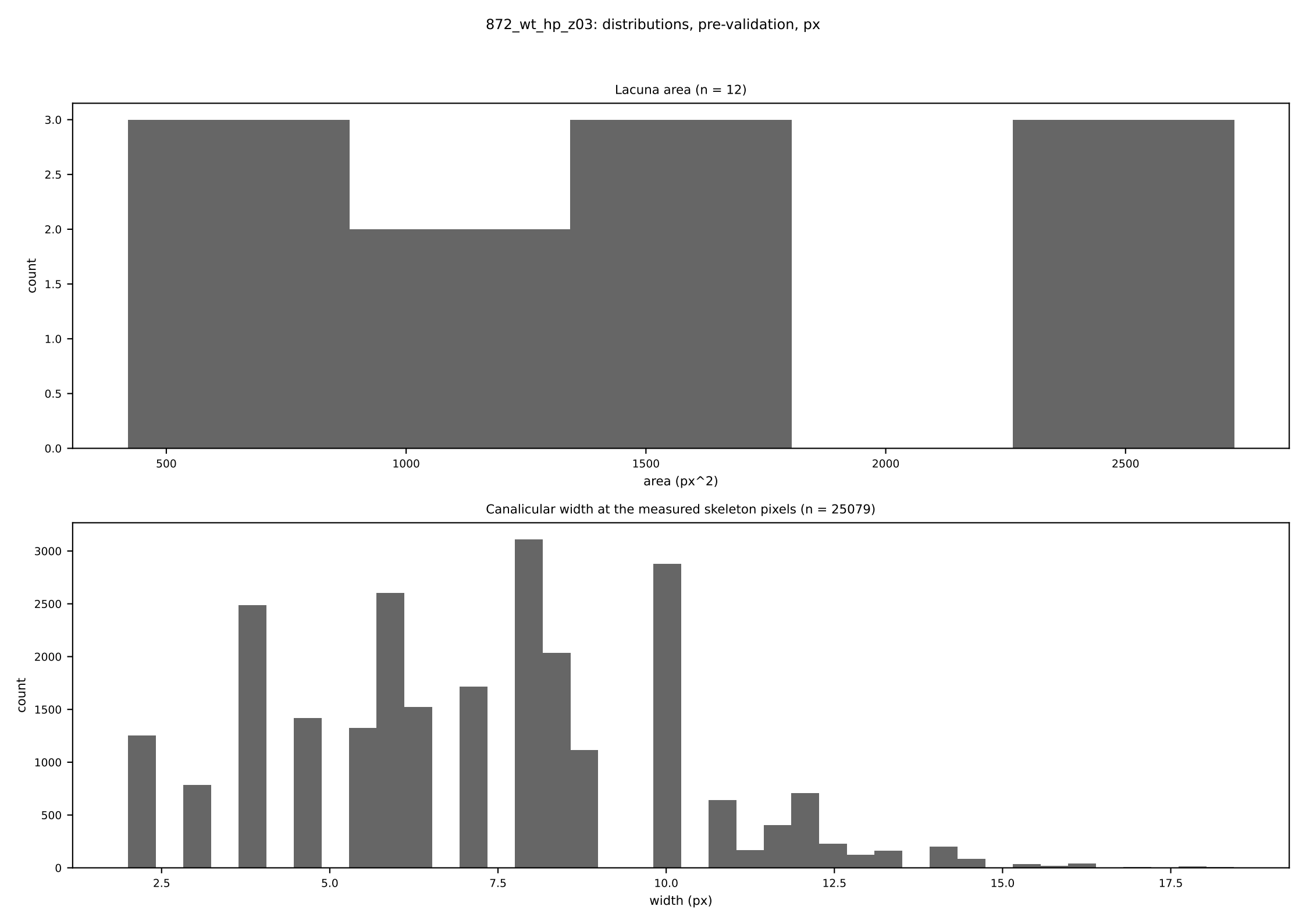
8	4	275	680	462.375	23	20.1032	6.7744
9	11	470	1100	4018.36	248	16.2031	8.3097
10	3	133	460	1165.26	61	19.1027	8.7365
11	14	525	1362	4422.98	285	15.5192	8.0788
12	3	219	697	2111.78	130	16.2445	8.0687

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	165.782	7287	20197	1	1	0.0332098	0.0263406	3.016
2	97.0538	5433	16006	1	1	0.0353396	0.0362989	6.1821
3	229.8	8292	22152	1	1	0.0277376	0.0280787	2.1758
4	283.664	6163	14859	0.6902	0.6384	0.0441344	0.0482536	2.1152
5	223.179	8740	22193	1	1	0.0356979	0.0407336	4.0326
6	253.279	8248	19209	0.8394	0.7603	0.0238846	0.0187412	1.5793
7	180.267	7666	20955	1	1	0.0530916	0.0545455	4.4379
8	192.551	7977	20442	1	1	0.0344741	0.0332648	2.0774
9	151.782	6950	19530	1	1	0.0676259	0.0563236	7.2472
10	92.7696	3398	9619	0.6214	0.5799	0.0391407	0.047822	3.2338
11	219.522	8093	21490	1	1	0.0648709	0.0633783	6.3775
12	65.8406	4580	14297	1	1	0.0478166	0.0487515	4.5565

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	199	247	265.267	589.416	6	4	8
2	129	149	226.179	679.11	5	5	8
3	193	318	256.652	681.044	5	4	8
4	222	589	303.693	815.411	7	9	7
5	237	495	351.007	1006.74	5	11	13
6	100	139	210.924	391.948	3	9	7
7	317	904	471.546	1301.77	7	10	15
8	259	465	316.907	790.394	7	6	11
9	467	1040	541.132	1250.13	9	14	16
10	129	392	152.054	517.919	3	3	6
11	517	1274	605.558	1570.15	10	11	19
12	186	400	250.309	796.453	4	7	8


872_wt_hp_z03: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 25079)
count
width (px)